



(a) Expert-Driven Curation

1. Sources

Top Journals, Conferences
& Popular databases

2. ML + Biology Experts

Data filtering, selection,
preprocessing, and splitting

3. Unified Public Task Capsules

Train/test data, task description, sample submission.

4. Hidden Private Labels

Ground truth labels, grading function.



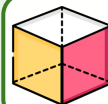
(b) 9 Biomedical Domains, 76 Tasks



Sequence



Single Cell



3D Structure



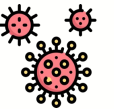
Network
Biology



Chemical
Biology



Perturbation
Dynamics



Phenotype
Disease



Imaging
Integrated



Text
Integrated



(c) Evaluating 11 Agent Configurations

4 Closed-source LLMs

GPT-5.4 Gemini-3.1-Pro
Claude-Opus-4.6
Qwen3.6-Plus

3 Open-source LLMs

GLM-5.1
Gemma-4-31B
DeepSeek-V3.2

2 Biomedical Agents

Biomni
STELLA

2 ML Coding Agents

MLEvolve
ML-Master 2.0



(d) Execution

Sandbox Runtime

2h wall-clock and 1 GPU
per task, unified conda
environment

Output

Submission.csv →
Held-out grader



(e) Metrics & Analysis

Evaluation Metrics (9 Types)

Pearson / Spearman / Accuracy
Macro-F1 / ROC-AUC / AUPRC
C-index / Dice / TM-score

Analysis Output

Leaderboard | Domain Heatmap
Failure Taxonomy | Cost Analysis